

MAGNITOGORSK, Russian Federation

WMO: 288380

Lat: **53.350N**Lon: **59.083E**Elev: **1261**StdP: **14.04**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.4	-16.8	-26.3	1.3	-21.1	-21.5	1.8	-16.4	22.6	19.4	20.1	20.7	2.3	10	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	87.5	65.1	84.2	64.0	81.2	62.8	68.0	81.3	66.4	79.5	64.8	77.1	8.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.4	91.5	73.5	61.7	86.2	71.9	60.0	81.0	70.7	33.0	81.3	31.7	79.8	30.5	77.2	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	21.7	18.5	16.2	-26.6	92.9	5.3	3.8	-30.5	95.6	-33.6	97.8	-36.6	99.9	-40.5	102.7	
(5)				-26.6	71.0	5.5	2.5	-30.6	72.8	-33.8	74.2	-37.0	75.6	-41.0	77.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	4.2	7.0	20.4	40.7	55.4	64.0	66.5	64.0	52.6	39.2	23.1	10.5	
	DBStd	24.41	11.31	11.36	9.59	9.80	8.38	7.68	6.28	7.16	8.02	8.56	11.47	12.20	
	HDD50	6336	1421	1204	919	306	42	5	0	2	61	344	808	1224	
	HDD65	10329	1886	1624	1384	728	311	108	58	106	378	799	1258	1689	
	CDD50	1762	0	0	0	28	210	425	513	436	140	10	0	0	
	CDD65	278	0	0	0	0	13	78	105	75	7	0	0	0	
	CDH74	3133	0	0	0	9	264	908	1008	824	120	0	0	0	
	CDH80	908	0	0	0	0	36	280	292	269	31	0	0	0	
Wind	WSAvg	6.8	5.4	6.5	7.3	8.1	7.6	7.0	6.9	6.2	6.1	7.0	6.9	6.2	
Precipitation	PrecAvg	14.4	0.6	0.5	0.6	0.9	1.2	1.9	3.0	2.0	1.1	1.1	0.8	0.8	
	PrecMax	20.0	2.4	2.3	1.8	3.0	3.5	4.6	6.7	7.2	4.4	4.2	2.2	1.8	
	PrecMin	5.8	0.1	0.0	0.0	0.0	0.1	0.0	0.3	0.2	0.1	0.0	0.1	0.0	
	PrecStd	3.2	0.5	0.5	0.4	0.6	0.9	1.1	1.6	1.4	0.8	0.7	0.5	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.5	34.7	48.2	74.6	83.9	91.2	91.8	90.9	84.1	68.4	50.0	36.5	
		MCWB	30.5	31.7	40.4	55.3	61.2	64.4	64.1	65.6	60.7	52.2	44.5	31.1	
	2%	DB	28.4	31.6	40.4	68.4	80.3	86.3	86.4	86.8	77.0	62.6	43.9	32.9	
		MCWB	26.7	29.0	34.7	52.2	59.1	64.4	65.4	65.3	58.1	49.6	40.1	30.6	
	5%	DB	25.4	27.9	36.1	63.0	77.0	83.1	83.0	82.6	72.0	57.4	40.1	30.2	
		MCWB	23.9	26.0	32.4	49.6	58.4	63.2	64.5	64.1	56.9	47.2	37.1	28.3	
	10%	DB	22.0	23.8	33.4	57.7	72.6	79.3	79.7	78.2	66.9	52.5	36.8	27.2	
		MCWB	20.8	21.9	30.9	46.7	56.2	62.2	63.4	62.8	54.9	45.0	34.1	25.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.0	32.2	41.4	56.9	64.7	68.9	70.0	70.5	63.4	54.3	45.4	33.1
			MCDB	32.5	34.1	47.4	71.1	78.1	83.6	82.9	83.9	79.7	64.8	48.6	35.1
2%		WB	27.0	29.2	35.4	53.1	61.3	66.6	68.0	67.8	60.0	51.0	40.6	30.8	
		MCDB	28.3	31.3	39.0	66.4	75.7	81.5	80.7	81.2	72.8	60.0	43.3	33.0	
5%		WB	24.0	26.2	32.9	50.3	59.4	65.0	66.5	65.8	57.8	48.2	37.3	28.3	
		MCDB	25.3	28.2	35.4	61.7	74.1	79.0	78.7	78.5	69.4	56.2	40.1	30.2	
10%		WB	20.9	22.0	31.2	47.4	57.3	63.4	64.9	63.9	55.6	45.6	34.5	25.7	
		MCDB	22.0	23.6	33.5	57.0	70.5	76.2	76.2	75.6	65.7	51.8	36.5	27.2	
Mean Daily Temperature Range	5% DB	MDBR	14.4	16.3	16.0	19.0	22.6	21.7	19.5	20.7	19.9	16.5	12.6	13.6	
		MCDBR	14.0	15.3	16.6	28.4	29.2	28.1	25.5	28.1	28.7	27.1	14.5	12.5	
		MCWBR	13.0	13.4	12.5	15.3	13.0	11.4	9.6	10.8	13.5	15.8	11.2	11.0	
	5% WB	MCDBR	13.2	14.2	15.6	26.5	26.6	24.5	21.6	24.2	25.2	24.3	13.6	11.9	
		MCWBR	12.4	12.8	12.5	14.9	13.1	11.4	9.6	11.0	12.7	15.0	11.2	10.7	
Clear-Sky Solar Irradiance	taub	0.276	0.303	0.358	0.407	0.396	0.390	0.411	0.429	0.372	0.340	0.304	0.273		
	taud	2.113	2.080	1.989	2.111	2.260	2.347	2.283	2.210	2.332	2.318	2.222	2.109		
	Ebn at Noon	214	246	254	259	270	273	264	250	252	234	203	188		
	Edh at Noon	22	32	43	44	40	37	39	40	31	25	20	18		
All-Sky Solar Radiation	RadAvg	333	679	1157	1452	1740	1901	1791	1465	1019	555	280	220		
	RadStd	28	48	89	128	150	161	120	140	103	95	36	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page